Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software engineer with 8+ years building production-grade backend systems and data pipelines, now specializing in generative AI. Built **AgentProbe** - an open-source generative AI evaluation framework with a 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination detection), CI-integrated regression gate, and multi-provider LLM routing - used to validate LLM and RAG pipeline quality across 1,240+ test cases. Built **Leasel**: an 8-agent LangGraph orchestration pipeline with hand-rolled RAG achieving 93.6% precision on 1,240 compliance documents, with context engineering and HITL escalation enforcing correctness at every regulated decision point. At Entrata, owned Python microservices, data pipelines, and Docker-containerized services at 100K+ MAU scale across 120+ Agile sprints with CI/CD. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). [AgentProbe: github.com/venkatalakshmiparimala/AgentProbe]

CORE COMPETENCIES

Generative AI Development | RAG Pipeline Engineering | AI Agent Orchestration (LangGraph) | LLM Evaluation Frameworks | Context Engineering | Python · LangChain · FastAPI | Docker · CI/CD · Agile Delivery | Production AI Microservices

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an agentic mission-planning system using LangGraph (parse → plan → retrieve → validate → conditional routing) - a generative AI pipeline translating natural-language survey requests into structured flight plans with compliance validation against FAA Part 107 regulations.
- Implemented a RAG layer retrieving from an FAA regulatory corpus, with each compliance check functioning as a grounded hypothesis test; engineered validation guardrails enforcing altitude limits and geometric no-fly-zone checks before MAVLink export.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant

- Delivered technical workshops to 600+ students across multiple sections covering Python programming (data structures, OOP, automation scripting) and database design (ER modeling, SQL); provided one-on-one debugging support and rubric-aligned grading.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; designed full pipeline from data collection and transformation through hypothesis testing and validation.
- Applied Isolation Forest to automate data integrity checks across 3 research pipelines, surfacing anomalies that manual review missed; designed Tableau and ggplot2 dashboards improving research comprehension 40%.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer

- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign with Docker containerization - scalable architecture delivering 50% throughput improvement and 30% deployment time reduction; implemented CI/CD pipelines eliminating 80% of manual operational workload.
- **Owned the annual HUD income-limit data pipeline for 7 years** - federal source rows transformed into ~180,000 domain records powering eligibility, rent, and tax-credit calculations for 100K+ housing units; every release included baseline validation, unit tests, reconciliation, and rollback preparation. Zero compliance violations across the ownership period.
- **Built RabbitMQ-based async processing system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees that eliminated double-processing failures; owned through production incidents.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant production schemas.
- **Led AWS S3 document migration** for Affordable Housing module - inventoried all document types, mapped legacy records, validated migrated content; implemented Libsodium-encrypted pipelines achieving 100% PHI compliance with zero security incidents.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built debugging tools reducing debugging time 40% and keeping bug count below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018 · Pune

Associate / Trainee Software Engineer

- Built property management modules (PHP, Python, PostgreSQL) for affordable housing compliance workflows; led cloud migration of client documents from legacy server storage to AWS S3, resolving 90% of legacy space constraints.
- Redesigned document storage database architecture, migrating hundreds of thousands of records to a modern schema; provided 6 months of post-deployment stabilization.

PROJECTS

AgentProbe: Generative AI Evaluation Framework

Open Source

2025 | Python, FastAPI, LangGraph, ChromaDB, CI/CD (GitHub Actions)

github.com/venkatalakshmiparimala/AgentProbe

Built a fully local, zero-cost generative AI evaluation framework: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate that blocks quality regressions before merge, and multi-provider LLM routing for dynamic tool selection. Designed for plug-and-play integration -- any agent connects via API; unit tests + LLM-as-judge scoring run automatically on every push. Used to validate LeaseI across 1,240+ compliance test cases: pass@1 81.7%, pass@3 88.3% via multi-trial sampling harness.

LeaseI: Agentic AI Investigation Platform

RAG + LangGraph

2026 | Python, LangGraph (8-agent), FastAPI, TF-IDF RAG, OCR, SQLite

github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an 8-agent LangGraph orchestration pipeline (Triage -> Retrieval -> Configuration -> Historical Incident -> Structured Data -> Reasoning -> Validation -> Human Escalation) achieving 93.6% precision and 87.9% F1 on 1,240 residential lease PDFs (5 US states + federal). Engineered a hand-rolled TF-IDF RAG layer with section-aware chunking, OCR fallback, and context engineering for accurate clause extraction; 100% citation correctness, 0% unsupported-claim rate on eval set. Core design enforced in code: LLM reasons, deterministic logic or human makes every final regulated decision.

AI on FHIR: NLP-Driven Medical Records Query System

2024 | Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings

github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records. Containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

Generative AI & Agents: LangChain, LangGraph, Generative AI, RAG Pipelines, Context Engineering, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Prompt Engineering, HITL, FastAPI, ChromaDB, Vector Databases, Semantic Search, Embeddings, NLP, OpenAI API, Anthropic API, MCP

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, PySpark / Apache Spark (Databricks-compatible), NumPy, Pandas, Tableau, Power BI, spaCy, BERT, Hugging Face

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum